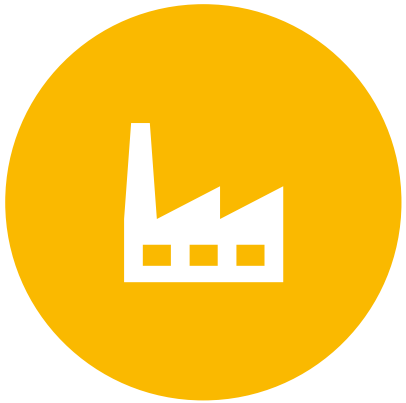
IND3001 Engineering Economics

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IMAGINE A FACTORY WORKER IN 1880. WHO DECIDES HOW FAST THEY WORK?



IMAGINE A FACTORY WORKER IN 1920. WHO DECIDES NOW?

Who decides HOW you do your job ?

Source. *P-O Börnfelt, Work Organisation in Practice: From Taylorism to Sustainable Work Organisations (Palgrave Macmillan, 2023), Chapter 2 "Taylorism and Fordism", pp. 25–37*

Before the factory: the artisan and the guild

- An **artisan** (craftsman) is a skilled worker who makes a whole product. The result depends completely on the craftsman's **skill and knowledge**, even if he uses machines and tools.
- Examples: shoemakers, weavers, bakers, goldsmiths, carpenters, butchers, saddle makers.

- Craftsmen belonged to a **guild**. A guild was an association (a kind of brotherhood) of craftsmen and merchants. Guilds existed in Europe at least from the early Middle Ages.

What did a guild do?

- **Social support:** helped members who were poor or sick, paid for funerals.
- **Quality control:** set rules for materials (for example, leather or wood). Each craftsman put his own mark on the product.
- **Training:** controlled who could become a craftsman. The path was: **apprentice** → **journeyman** → **master.** Training took a long time — often about **ten years**.
- **Solving disputes** inside the trade.
- **Coordinating the economy:** for example, in Bologna the tanners' guild and the shoemakers' guild negotiated the price and quality of leather. This created a supply chain.

Were guilds bad for progress?

Many European countries closed them in the 18th century.

There were **many innovations before the Industrial Revolution**

New ideas spread very fast, because there was no patent system.

Most innovations were **small, step-by-step improvements**, discovered while doing the work.

Craftsmen were **very mobile**.

Important point for engineers: skilled craftsmen were the **ancestors of the engineering profession**.

The move to bigger workshops

- In the early industrial period, craftsmen started to work in **larger groups** and **bigger workshops**. Someone had to manage and coordinate the work. The **employer (the owner of capital)** wanted to take that role.
- Step by step, work became more controlled:

-fixed working hours,

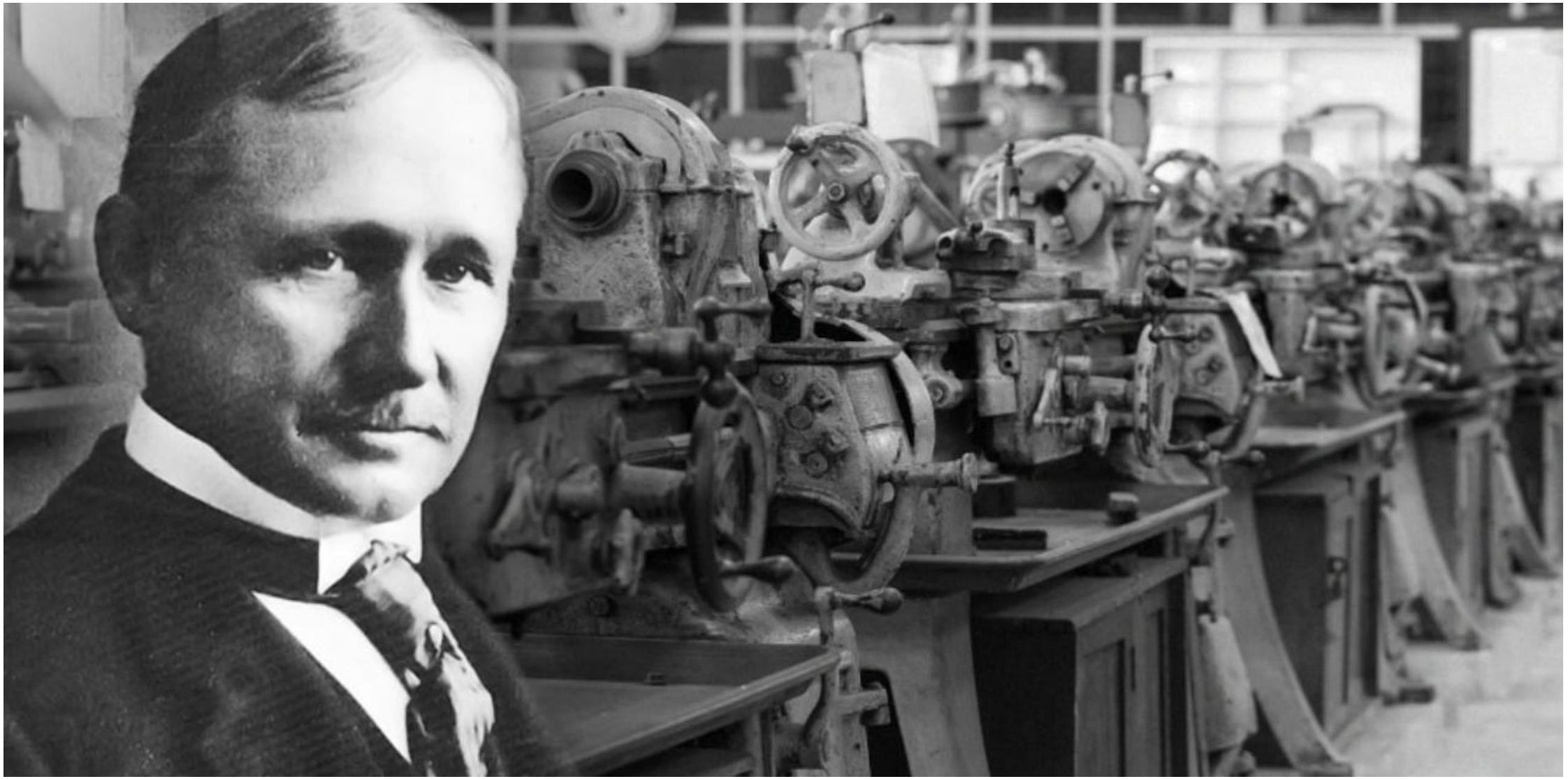
-foremen who watched the workers,

-rules,

-minimum production levels.

In the artisan world, the WORKER owned the knowledge. That was about to change.

FREDERICK WINSLOW TAYLOR (1856–1915) AND SCIENTIFIC MANAGEMENT



Taylor's problem: why are workers slow?

Taylor's main goal was to **increase efficiency**.
He saw two reasons for low efficiency:

Reason 1 — Workers work slowly on purpose.

- **"systematic soldiering"**
- workers made **informal agreements** among themselves about a "reasonable" production level. Anyone who produced more felt **group pressure**.
- They were paid by the piece (**piecework**)

Reason 2 — Work methods are unsystematic.

- The **workers knew more about production than their bosses.** They decided how fast to run the machines. If a new foreman tried to speed things up, workers resisted.

Taylor's solution: four main ideas

- **Idea 1 — Management takes control of the work process.**
- Rules alone are not enough while workers still control production. Management must study the work in detail, find the **"one best way"** to do each task, and **standardise** it. Planning moves to a **planning department** run by **white-collar workers**.

- **Idea 2 — Work is organised individually.**
- To break the workers' informal agreements, each worker works **alone**, trained by a supervisor, and cooperates only with the supervisor. Workers are tested and selected. Those who do not meet the standard are trained or **dismissed**.

- **Idea 3 — Pay is tied to the standard.**
- If a worker reaches the daily target, he gets the full rate. If not, he gets **less pay**. After reorganisation, pay should be **30–100% higher**, so that workers accept the faster pace. Taylor talked about "friendly relations" between management and workers.

- **Idea 4 — White-collar experts plan the work in detail** using **time and motion studies**

According to Taylor, one of the most important success factors was to obtain the attitude changes of workers. A worker should not see the employer as an opponent who wanted to exploit the worker as much as possible. Instead they should be seen as friends working side by side towards the same goal (Taylor, 1947/1903, pp. 49–56)

Example: loading pig iron

	Before Taylor	After Taylor
Pig iron moved per worker per day	12–13 tons	45–48 tons
Work organisation	Groups of 5–20 men	Individual work
Pay per day	about $1.15	$1.85 (+60%)
Who could do it?	Most workers	Only 1 in 8 men

Labour cost per ton:
Before: $1.16 ÷ 12.5 tons ≈ **$0.093 per ton**
After: $1.85 ÷ 47 tons ≈ **$0.039 per ton**

So the **labour cost per ton fell by about 58%,** even though the daily wage went **up by 60%**.

Time and motion studies

- chose some of the best workers,
- watched how they worked and what tools they used,
- measured **how long each step took**,
- removed **unnecessary motions**,
- tested tools (for example, **different shovels for different materials**),
- combined the best method + best tool into a **standard task**.

Methods-Time Measurement

Frank B. Gilbreth created a **general system**:

- He split all work into **basic body movements,**
- Each movement got a **standard time**.
- This system is called **MTM (Methods-Time Measurement)**.

With MTM, planners can calculate **how much time and how many workers** a job needs — **without watching the workers**.

Control by functional leadership

On the shop floor	In the planning department
Gang boss — shows how to use the machines in the most efficient way	Order of work / route clerk — plans the order of work and how products move
Speed boss — checks that machines run at the right speed	Instruction card clerk — writes the instruction card: drawings, tools, exact method, time allowed, pay
Inspector — checks quality and work speed	Time and cost clerk — records time and cost, calculates wages
Repair boss — checks that workers maintain the machines	Shop disciplinarian — punishes workers who disobey, are late or absent

HENRY FORD AND MASS PRODUCTION





Taylor's principles + mass production + the assembly line

FORD'S MASS PRODUCTION

The main futures of Fordism:

- Production of **a few, highly specialised products**.

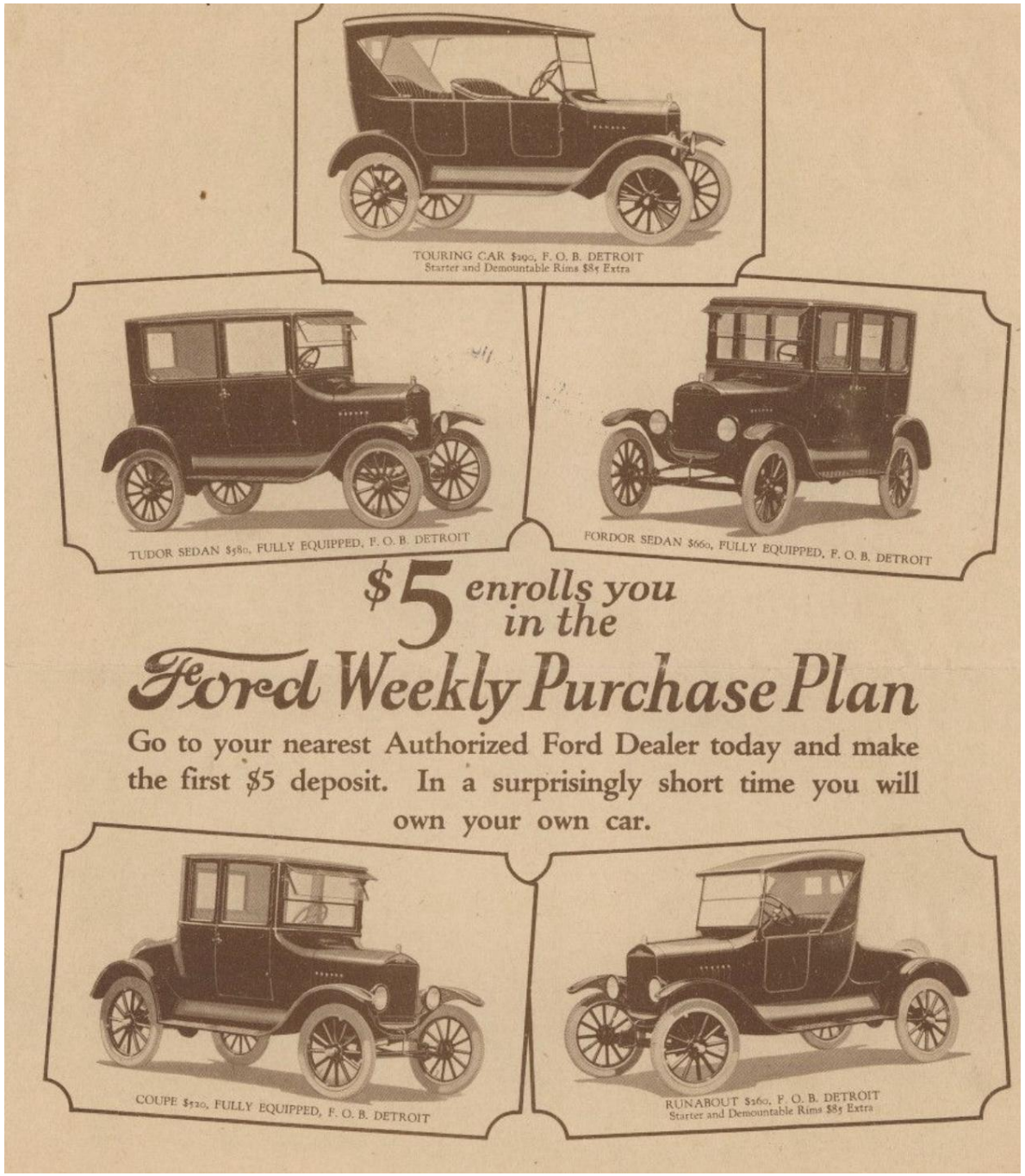
- Specialised machines.

- The **assembly line** — the **machine sets the pace** of work.

- Standardised, repetitive, short-cycle tasks — often only **1–2 minutes** per cycle. Workers have almost **no autonomy**.

-Strong hierarchy and **functional organisation**: planning, control and quality control are done in special departments.

The story of the Model T



How did Fordism spread?

- After World War I, and even more after World War II, **trade unions accepted monotonous work in exchange for higher wages.**

- Between the two world wars, Fordism spread across North America.
- **Europe** adopted mass production mainly in the **1950s and 1960s**.

- In the late 1960s and early 1970s there were **worker revolts in Europe** against this kind of work. This led to new models, such as the **sociotechnical model.**

- In the 1980s the USA adopted **Japanese models** (lean production) — but these were **also based on Fordism**. Europe followed in the 1990s.

Critique: what did workers lose?

Harry Braverman (*Labor and Monopoly Capital*, 1974) argued that work is **degraded** because **all the thinking is taken away from the worker**.

Management (white-collar)	Worker (blue-collar)
brain	hand
planning	making
concept	execution

Robert Blauner (1964) showed that repetitive, fragmented work, where the worker has no power, creates **alienation**. The worker becomes "a cog in a machine". He named four types:

Type	Simple meaning
Powerlessness	Others (or a system) control you.
Meaninglessness	You do one small part and cannot see the whole product or the goal.
Isolation	You are not part of a social group.
Self-estrangement	Your work has no link to your identity, your future or your life outside work.

Thanks!
Q&A

